

Civil Engineering

Vectors, Matrices and Data Analysis

Learner Notes

Vector operations, matrices, simultaneous equations, statistics, and engineering data.

Learning outcome	Focus
Apply vector operations	Applied mathematics in workplace and training contexts.
Use matrices to solve simultaneous equations	Applied mathematics in workplace and training contexts.
Analyze engineering data	Applied mathematics in workplace and training contexts.

1. Topic Overview

This topic develops mathematical skills required in Civil Engineering. Learners should show clear method, correct units, and technical interpretation.

2. Key Formulas

$$\vec{R} = \vec{A} + \vec{B}$$

$$AX = B$$

$$\det(A) = ad - bc$$

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

3. Worked-Example Method

Step 1: identify the given data. Step 2: select the correct formula. Step 3: substitute values carefully. Step 4: compute. Step 5: interpret the answer with correct units.

$$\vec{R} = \vec{A} + \vec{B}$$

4. Practice Questions

1. Resolve forces into components.

2. Solve simultaneous equations using matrices.
3. Calculate standard deviation of concrete test results.